

Fate mapping reveals origins and dynamics of monocytes and tissue macrophages under homeostasis.

SCIFACT-EVIDENCE-7521113

Evidence abstract

Mononuclear phagocytes, including monocytes, macrophages, and dendritic cells, contribute to tissue integrity as well as to innate and adaptive immune defense.

Emerging evidence for labor division indicates that manipulation of these cells could bear therapeutic potential.

However, specific ontogenies of individual populations and the overall functional organization of this cellular network are not well defined.

Here we report a fate-mapping study of the murine monocyte and macrophage compartment taking advantage of constitutive and conditional CX(3)CR1 promoter-driven Cre recombinase expression.

We have demonstrated that major tissue-resident macrophage populations, including liver Kupffer cells and lung alveolar, splenic, and peritoneal macrophages, are established prior to birth and maintain themselves subsequently during adulthood independent of replenishment by blood monocytes.

Furthermore, we have established that short-lived Ly6C(+) monocytes constitute obligatory steady-state precursors of blood-resident Ly6C(-) cells and that the abundance of Ly6C(+) blood monocytes dynamically controls the circulation lifespan of their progeny.

Document metadata

Corpus | SciFact

Document ID | 7521113

Evidence checklist

1. Mononuclear phagocytes, including monocytes, macrophages, and dendritic cells, contribute to tissue integrity as well as to innate and adaptive immune defense.
2. Emerging evidence for labor division indicates that manipulation of these cells could bear therapeutic potential.
3. However, specific ontogenies of individual populations and the overall functional organization of this cellular network are not well defined.
4. Here we report a fate-mapping study of the murine monocyte and macrophage compartment taking advantage of constitutive and conditional CX(3)CR1 promoter-driven Cre recombinase expression.
5. We have demonstrated that major tissue-resident macrophage populations, including liver Kupffer cells and lung alveolar, splenic, and peritoneal macrophages, are established prior to birth and maintain themselves subsequently during adulthood independent of replenishment by blood monocytes.
6. Furthermore, we have established that short-lived Ly6C(+) monocytes constitute obligatory steady-state precursors of blood-resident Ly6C(-) cells and that the abundance of Ly6C(+) blood monocytes dynamically controls the circulation lifespan of their progeny.

Source continuity

The canonical evidence above remains the authoritative retrieval target.